

Quiz 3 - Computational Physics I

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SCORE: 18.8 / 20

Date: Wednesday 30 April 2025

Duration: 45 minutes

Credits: 20 points (5 questions)

Type of evaluation: LAB

Provide concise answers to the following items:

1. (4 points) Numerical differentiation

- (a) Indicate the sources of errors when computing the divergence of a 2D magnetic field: $\vec{\nabla} \cdot \vec{B}$
- (b) What defines the order of accuracy of a finite difference method?

(a) The floating point precision errors and the truncation error of the Taylor series to carry out the finite differences. In the 2D divergence, we have both of these errors in x and y .

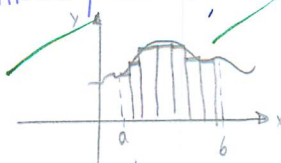
(b) The truncation error of the Taylor series and it we're using forward, backward or central finite differences.

2. (4 points) Numerical integration

- (a) Indicate 2 numerical methods used for calculating 1D integrals numerically.
- (b) Briefly explain how each method works.

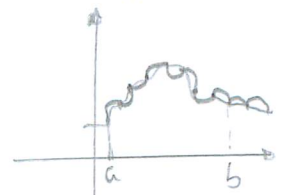
Riemann Sums

We approximate the area under the curve with rectangles. Then, we sum the area of each rectangle. Note that we can use the left, right or middle point to get the height.

$$\int_a^b f(x) dx \approx \sum_{n=0}^{N-1} h f(x_n), \quad h = x_{i+1} - x_i$$


Simpson's rule

We interpolate a quadratic polynomial between two data points. Then, we easily integrate the polynomial.



3. (4 points) Root finding in python

- (a) List 2 methods that we can use to find the roots of 1D functions in python.
- (b) Briefly explain how these python methods work to obtain the roots.

Numpy

`np.roots([a, b, c, d, ...])`; $a, b, c, d, \dots$ are the coefficients of a polynomial of degree n .

Newton-Raphson method

We compute the derivative of the function at a point, then we find the intersection of the tangent line with the x -axis. We use this new point to repeat the process. Generally, it will converge provided that $f'(x_n) \neq 0$ always.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

4. (4 points) Fourier transforms

- (a) Provide a concise definition of the Fourier Transform and explain its significance for physics.
 (b) Write down the mathematical formulae for the continuous and discrete Fourier Transforms.

A Fourier transform is a mathematical method that converts a N-dimensional signal from its real space (often time or space) to its (temporal or spatial) frequency domain, we call it Fourier space. It's very important in physics since we get discrete data from the experiments. They're, most of the time, polluted with noise or something else we don't want. Thus, by carrying out a Fourier transform of the data, we're able to ~~manipulate~~ ^{study and filter} the frequencies (filtering or selecting amplitudes above noise level), which we could not have done in real space.

$$\text{F.T: } F(k) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi kx} dx, k \in \mathbb{R} \quad \text{D.F.T: } X(k) = \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k \frac{n}{N}}, k \in [0, N]$$

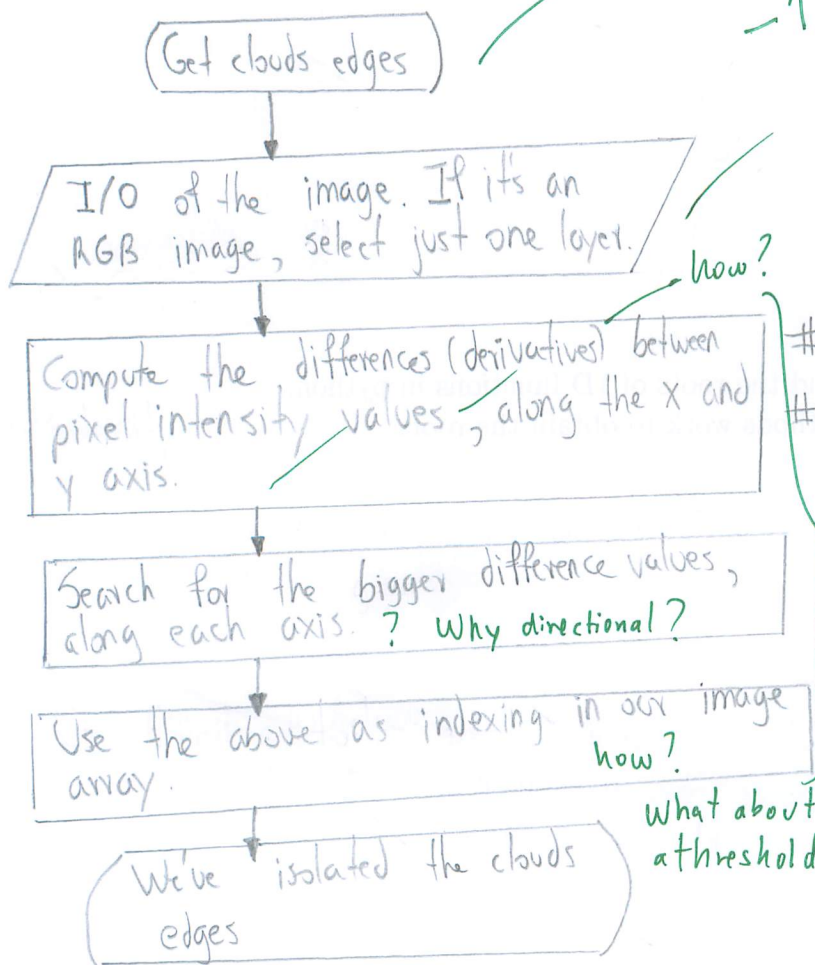
N is the size of the data

5. (4 points) Image processing: edge detection



Imagine you obtain the photograph (shown on the left-hand side) of atmospheric clouds, and you are asked to find and isolate the edges of the clouds.

Design and sketch a suitable algorithm workflow to achieve this goal in python.



-1.25

(sorry for the "bonón", my eraser was not cooperating).

np.gradient (array, stepsize, axis)
 # It's very important to define the step size.
 $dx = x[1] - x[0]$, for example

Organise this in functions and calls.

We select just those values matching the greatest differences.
 # We can do this since np.gradient() returns an array, along each axis, the same size as the original.

What about a threshold?